

The modulus function

Lessons 15–16 · Algebra 11, §1.5 (extension) · P1 · 1.6 · P2 · 1.1–1.3 · 28 problems

A modulus equation splits into two cases; a modulus inequality is a region. Check every root against the original equation — squaring and case-splitting both invent answers.

Easy 7

Reading the definition. $|x| = a$ has two answers when $a > 0$; $|x| \leq a$ is the interval between them.

1 $|7 - 12| = 5$

The bars come off after the subtraction: $|-5| = 5$.

2 $|-3| + |-4| = 7$

$3 + 4$. Note that this is not $|-3 - 4| = 7$ by accident — see problem 22.

3 Solve $|x| = 11$ $x = \pm 11$

Two numbers are 11 units from zero.

4 Solve $|x + 5| = 2$ $x = -3$ or $x = -7$

$x + 5 = 2$ gives $x = -3$; $x + 5 = -2$ gives $x = -7$.

5 Solve $|x| \leq 4$ $-4 \leq x \leq 4$

Distance from zero at most 4, so a closed interval.

6 Solve $|x| > 7$ $x < -7$ or $x > 7$

Distance from zero more than 7 — two separate pieces, never one.

7 Give the vertex of $y = |x + 3| - 2$ $(-3, -2)$

The corner sits where the inside is zero, at $x = -3$, and the -2 lowers it.

Medium 7

The inside is now linear in x , so each case still gives one root. In 12 both sides carry a modulus, which gives two cases and no more.

8 Solve $|3x - 2| = 10$ $x = 4$ or $x = -\frac{8}{3}$

$3x - 2 = 10$ gives $x = 4$; $3x - 2 = -10$ gives $3x = -8$.

9 Solve $|5 - 2x| = 1$ $x = 2$ or $x = 3$

$5 - 2x = 1$ gives $x = 2$; $5 - 2x = -1$ gives $x = 3$.

10 Solve $|x - 4| \leq 6$ $-2 \leq x \leq 10$

Read it as a distance: x is within 6 of 4, so $4 - 6 \leq x \leq 4 + 6$.

11 Solve $|4x + 1| \geq 9$ $\left| x \leq -\frac{5}{2} \text{ or } x \geq 2 \right|$

$4x + 1 \geq 9$ gives $x \geq 2$; $4x + 1 \leq -9$ gives $4x \leq -10$.

12 Solve $|x + 2| = |3x - 4|$ $\left| x = 3 \text{ or } x = \frac{1}{2} \right|$

Two moduli give two cases only. $x + 2 = 3x - 4$ gives $x = 3$; $x + 2 = -(3x - 4)$ gives $4x = 2$.

13 Give the vertex and the y-intercept of $y = |2x - 6|$ $\left| (3, 0); 6 \right|$

The corner is where $2x - 6 = 0$. At $x = 0$, $y = |-6| = 6$.

14 Give the range of $y = 5 - |x - 1|$ $\left| y \leq 5 \right|$

$|x - 1| \geq 0$, so subtracting it can only lower 5. The greatest value is 5, reached at $x = 1$.

Hard 7

Now the cases have to be checked. In 15 one case produces a root that the original equation rejects; in 17 the middle interval produces no root at all.

15 Solve $|x + 1| = 3x - 5$ $\left| x = 3 \text{ only} \right|$

$x + 1 = 3x - 5$ gives $x = 3$; check: $|4| = 4$ and $3(3) - 5 = 4$. ✓ $x + 1 = -(3x - 5)$ gives $4x = 4$, so $x = 1$; check: $|2| = 2$ but $3(1) - 5 = -2$. ✗ A modulus can never equal a negative number.

16 Solve $|x^2 - 9| = 7$ $\left| x = \pm 4 \text{ and } x = \pm \sqrt{2} \right|$

$x^2 - 9 = 7$ gives $x^2 = 16$; $x^2 - 9 = -7$ gives $x^2 = 2$. Four roots, because the inside is quadratic.

17 Solve $|x - 5| + |x + 2| = 9$ $\left| x = -3 \text{ or } x = 6 \right|$

The two corners are at $x = -2$ and $x = 5$, which split the line into three parts. For $x < -2$: $(5 - x) + (-x - 2) = 3 - 2x = 9$, so $x = -3$. ✓ For $-2 \leq x \leq 5$: the sum is always 7, never 9. For $x > 5$: $2x - 3 = 9$, so $x = 6$. ✓

18 Solve $|2x - 1| < |x + 3|$ $\left| -\frac{2}{3} < x < 4 \right|$

Both sides are non-negative, so squaring is safe here: $4x^2 - 4x + 1 < x^2 + 6x + 9$, that is $3x^2 - 10x - 8 < 0$, or $(3x + 2)(x - 4) < 0$. A negative product means x lies between the roots.

19 Solve $|x| - x = 8$ $\left| x = -4 \right|$

For $x \geq 0$ the left side is 0, never 8. For $x < 0$ it is $-2x = 8$.

20 The graph of $y = |x - a|$ passes through $(1, 4)$ and $(7, 2)$. Find a . $\left| a = 5 \right|$

$|1 - a| = 4$ gives $a = 5$ or $a = -3$; $|7 - a| = 2$ gives $a = 5$ or $a = 9$. Only $a = 5$ satisfies both.

21 Solve $|x^2 - 2x| = x$ $\left| x = 0, x = 1, x = 3 \right|$

$x^2 - 2x = x$ gives $x(x - 3) = 0$; $x^2 - 2x = -x$ gives $x(x - 1) = 0$. The right side must be non-negative, and all three roots satisfy $x \geq 0$, so all three stand.

Very hard 7

Two corners, or letters instead of numbers. A sketch of each piece is faster than any amount of case algebra.

- 22 Show that $|a| + |b| \geq |a + b|$, and say when the two sides are equal.

Equal when a and b have the same sign, or one of them is 0

Squaring both sides, the claim becomes $a^2 + 2|ab| + b^2 \geq a^2 + 2ab + b^2$, that is $|ab| \geq ab$, which is true for every real number. Equality needs $|ab| = ab$, so $ab \geq 0$ — the two numbers point the same way. This is the triangle inequality.

- 23 Solve $|x - 1| + |x - 3| < 4$. $0 < x < 4$

Corners at 1 and 3. For $x < 1$: $4 - 2x < 4$, so $x > 0$, giving $0 < x < 1$. For $1 \leq x \leq 3$: the sum is 2, which is always less than 4. For $x > 3$: $2x - 4 < 4$, so $x < 4$. The three pieces join into one interval.

- 24 Find the greatest and least values of $y = |x - 2| + |x + 4|$. least 6; no greatest value

Between the corners, for $-4 \leq x \leq 2$, the sum is $(2 - x) + (x + 4) = 6$ — constant, and that constant is the distance between -4 and 2. Outside, both terms grow, so $y \rightarrow \infty$. The minimum is reached on a whole interval, not at one point.

- 25 Solve $|x + 3| \geq |2x|$. $-1 \leq x \leq 3$

Both sides non-negative, so square: $x^2 + 6x + 9 \geq 4x^2$, that is $3x^2 - 6x - 9 \leq 0$, or $(x - 3)(x + 1) \leq 0$.

- 26 Explain why $y = |x^2 - 1|$ has no derivative at $x = 1$ but does have one at $x = 0$. a corner at $x = 1$; $y'(0) = 0$

For $x > 1$ the inside is positive, so $y = x^2 - 1$ and $y' = 2x \rightarrow 2$. For x just below 1 the inside is negative, so $y = 1 - x^2$ and $y' = -2x \rightarrow -2$. The one-sided gradients differ, so there is no tangent. Near $x = 0$ the inside is negative throughout, so $y = 1 - x^2$ with no switch at all, and $y'(0) = 0$.

- 27 $|x - 2| = k$ has exactly two solutions. Find the values of k . $k > 0$

The graph of $y = |x - 2|$ is a V with its corner on the x -axis. A horizontal line $y = k$ cuts it twice when $k > 0$, once when $k = 0$ and never when $k < 0$.

- 28 Sketch $y = |x| - |x - 4|$ and give its range. $-4 \leq y \leq 4$

Three pieces. For $x < 0$: $(-x) - (4 - x) = -4$. For $0 \leq x \leq 4$: $x - (4 - x) = 2x - 4$, a line from -4 to 4. For $x > 4$: $x - (x - 4) = 4$. The graph is flat, then a ramp, then flat again.